RetroCalc User Manual

Historical Energy Analysis and Level 2 Energy Modeling for Multifamily Buildings

NEW ECOLOGY
Community-Based Sustainable Development

About this manual. This guide is written for those using RetroCalc as a modeling tool. It assumes no programming experience. **Part 1 (Getting Started)** gives you a fast overview of the app's sections and how a project flows through them. **Part 2 (Reference)** explains each input in detail and how it drives the underlying energy calculations. Appendices at the end cover scripted/batch use, the Excel input reference, and field data-collection checklists.

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Part 1 — Getting Started

1. Overview

RetroCalc, developed by New Ecology Inc. (NEI), is an innovative energy modeling tool designed specifically for decarbonizing multifamily buildings at scale. Drawing on NEI's 25 years of experience in multifamily retrofits, RetroCalc combines detailed energy and water modeling with an intuitive, user-friendly functionality. This tool provides technical assistance (TA) providers with the ASHRAE Level II-compliant energy and cost analysis that is needed to achieve successful decarbonization, without the time and complexity typically required by traditional modeling software.

This tool can be used for a variety of different project scopes, from basic utility data analysis to modeling the impacts of Energy Conservation Measures (ECMs) in a building retrofit scope. Since this work is rooted in an existing building, the efficacy of the model depends on quality data collection by TA providers, which is the first step (Figure 1) of using RetroCalc.



Figure 1: Process Flow

- **Historical Energy Analysis (HEA)** — a top-down picture built from the building's **actual utility bills**. RetroCalc reads your monthly electricity, gas, water, oil, and propane records and statistically separates total consumption into end uses (base load, space heating, space cooling, hot water, cooking/drying). It then weather-normalizes the result so different years can be compared on equal footing.
- **Building the ASHRAE Level 2 (L2) model** — a bottom-up engineering picture built from the building's **physical characteristics**: geometry, envelope, HVAC, hot water,

appliances, and ventilation. RetroCalc runs a steady-state monthly energy balance to estimate the same end uses from first principles.

- **Calibration** — RetroCalc generates visualizations to help you compare the HEA to the L2 model. Edit parameters on the L2 model and recalculate — all on the same screen — to quickly calibrate to the HEA.
- **Define ECMs and Packages** — Add ECMs that map directly to L2 model parameters to define the changes relevant to your retrofit. Package those ECMs together in one or more scenarios to run against the baseline building model.
- **Estimate Savings** — Apply the packages you created to the building to model the changes. Mix and match and re-run to do scenario planning and choose the packages that best fit your project goals. RetroCalc reports the cumulative energy, cost, and carbon impact of each package against the existing baseline and other packages.

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2. Key Concepts and Vocabulary

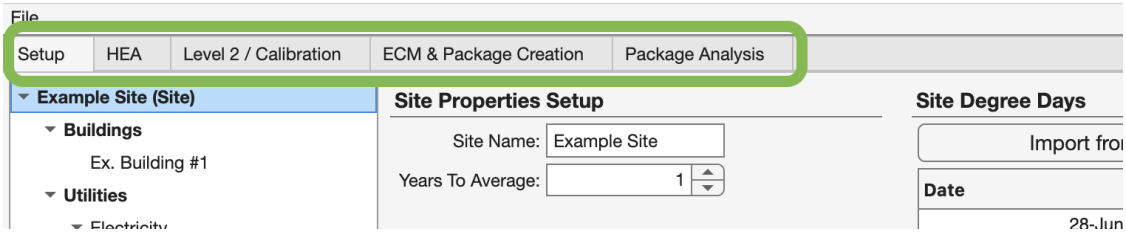
Term	Meaning
HEA (Historical Energy Analysis)	The bill-based analysis. Characterizes actual past energy and water use and splits it into end uses.
Level 2	The physics-based model built from building inputs. Named "Level 2" to distinguish it from the simpler bill-only "Level 1" view.
End use	A category of consumption: base/plug load, space heating, space cooling, domestic hot water (DHW), cooking, and clothes drying.
Baseline	The building as it exists today, modeled in Level 2 with no measures applied. All savings are reported relative to this.
ECM (Energy Conservation Measure)	A single improvement (e.g., higher-efficiency boiler, added wall insulation, low-flow fixtures).
Package	An ordered group of ECMs applied together. Each ECM builds on the previous one so you see cumulative impact.
EUI (Energy Use Intensity)	Total annual energy divided by conditioned floor area, in kBtu/ft^2/year. The headline comparison metric.
HDD / CDD (Heating / Cooling Degree Days)	Weather measures of how far and how long outdoor temperature is below 65 °F (HDD) or above 70 °F (CDD). Higher HDD → more heating; higher CDD → more cooling.
DHW (Domestic Hot Water)	Hot water for showers, faucets, and appliances — separate from space heating.

Term	Meaning
Infiltration	Uncontrolled air leakage through cracks and gaps in the envelope, driven pressure and temperature difference.
Ventilation	Deliberate, mechanical outdoor-air supply for indoor air quality — distinct from infiltration.
Weather normalization	Adjusting a year's heating/cooling energy to a typical weather year, so multi-year comparisons aren't distorted by an unusually mild or harsh season.

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3. The Workflow at a Glance

RetroCalc is organized into **five sections**, worked left to right. Each section feeds the next.



#	Section	What you do here
1	**Setup**	Define the site, the building(s), and the utility meters. Enter or import billing and weather data. To know more about how to import refer to Appendix A.
2	**HEA**	Run the bill-based analysis and review actual end-use breakdowns and costs. Refer to Appendix B for more.
3	**Level 2**	Enter the building's physical details, run the physics model, and calibrate it against the HEA.
4	**ECM Package**	Build a library of measures (ECMs) and assemble them into packages.
5	**Package Analysis**	Apply packages to a building and review cumulative energy, cost, and carbon savings.

The short version of a project:

1. In **Setup**, enter the site location and building characteristics, add each utility meter, and import the bills and the historical weather and the typical meteorological weather (degree-day) file.
2. In **HEA**, run the analysis. Review the end-use breakdown and confirm it looks reasonable.

3. In **Level 2**, enter envelope, HVAC, DHW, appliance, and ventilation details. Run the model and compare it to the HEA, adjusting inputs until the two agree.
4. In **ECM Package**, define your retrofit measures and group them into a package.
5. In **Package Analysis**, assign the package to the building and read off the savings.

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4. Preparing Your Input Files

RetroCalc's inputs are entered through the app's UI, so the work that determines how quickly a project goes is gathering good data before you begin entering it. This section covers how to do that, roughly in the order you'll need it:

1. **Contact the building owner early.** Ask for building plans and utility bills, and send the utility data collection workbook so bills come back in a consistent, importable format. At the same time, ask whether the owner already has retrofit plans or specific ECMs in mind — that scope shapes what you'll focus on during the site visit. Appendix B has the full request checklist.
2. **Run the HEA before your site visit.** Once utility bills are obtained, run the Historical Energy Analysis first. The end-use breakdown shows which loads — heating, cooling, DHW, base load — are largest, so you know what to prioritize during a limited site visit.
3. **Collect on-site data for the Level 2 model.** Walk the building and record the physical characteristics the Level 2 model needs: envelope, HVAC, DHW, appliances, and ventilation. Appendix C has a system-by-system checklist organized the same way as Section 8.

The historical degree days data can be pulled from an inbuilt API, pulled from degreedays.net. For information about how to set-up look at Appendix A. Once you have this data, it's loaded into RetroCalc through three kinds of Excel workbook (`.xlsx`): a site input workbook, a utility usage workbook, and a historical degree-day file. Exact sheet names, cell layouts, and required columns for each are listed in Appendix A.

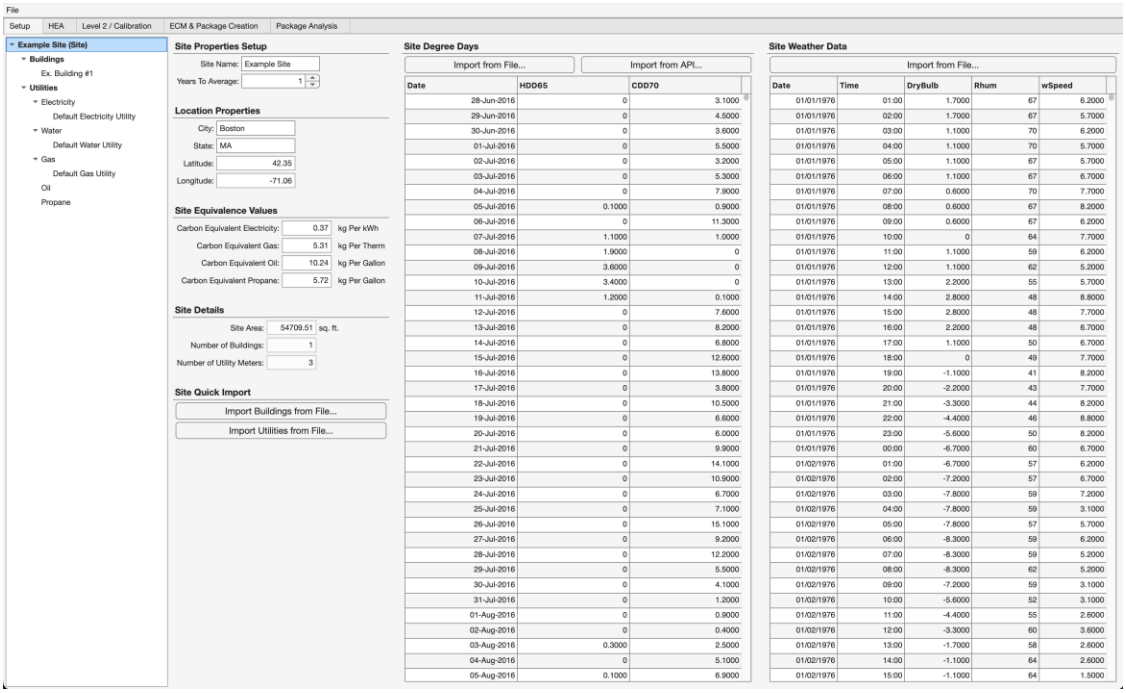
> **Tip.** The historical degree-day history should cover at least the number of past years you intend to average for weather normalization (set in Setup), and should extend through the last year of billing data. It can be hand-compiled from a nearby weather station's daily records, or pulled from a public degree-day data service if one is available for your site — either way it just needs to match the format in Appendix A.
>
> **Tip.** Separately from the historical degree-day file, RetroCalc also needs a TMY (Typical Meteorological Year) or similar representative weather file for the site, used for the solar and ground-temperature lookups described in 6.1 Site.

5. The Five Sections, Step by Step

5.1 Setup

Setup is where you describe the project. It has three parts:

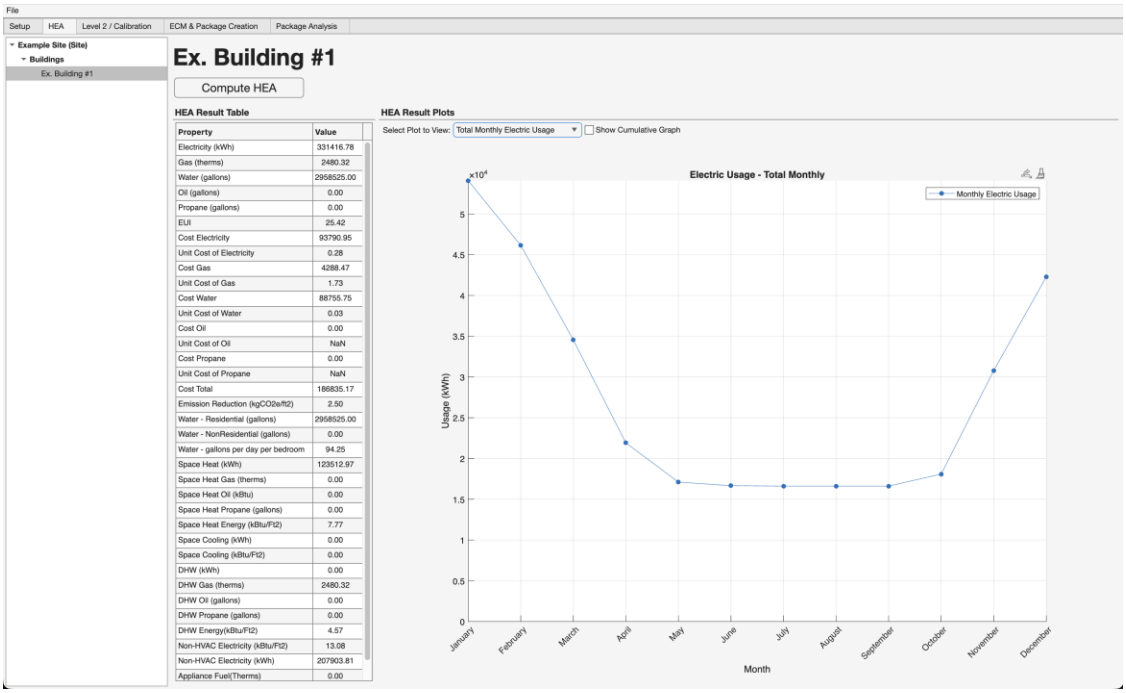
- **Site** — project name, location (city, state, and latitude/longitude), per-fuel carbon factors, the shoulder months used in the bill analysis, and how many past weather years to average for normalization.
- **Buildings** — add one or more buildings and enter each building's physical characteristics (areas, units, bedroom mix, year built, stories, thermal-mass class, heating/cooling season dates, setpoints).
- **Utility meters** — add each electricity, gas, water, oil, or propane meter, import its monthly billing data, and set its **end-use flags** (which loads that meter serves).



5.2 HEA

The HEA section runs the bill-based analysis and displays the results. Running it processes every meter's billing data, separates it into end uses, weather-normalizes the seasonal portion, and rolls the meters up to each building and to the whole site.

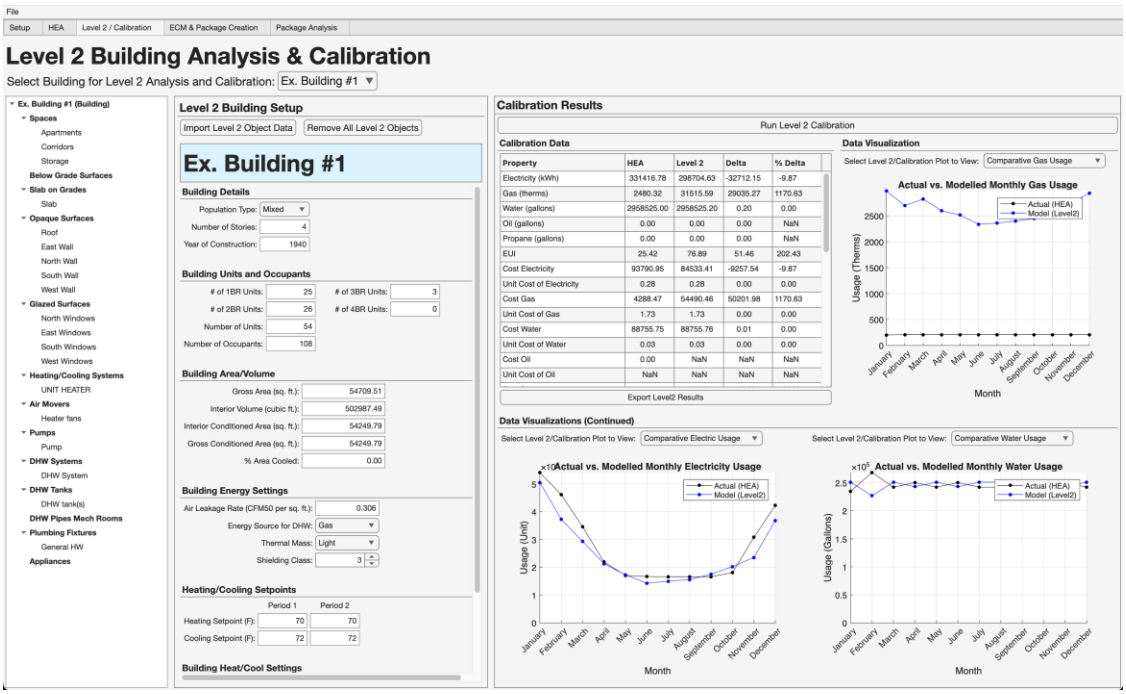
You review per-building results — energy use intensity, annual usage and cost by fuel, carbon, and the end-use breakdown — plus a site-level summary that combines all buildings by area.



5.3 Level 2

The Level 2 section is where you build the engineering model. You enter detailed component data for the building — heating and cooling systems, envelope surfaces, hot-water equipment, spaces, appliances, plumbing fixtures, ventilation, and pumps — then run the model to compute modeled energy use.

A **calibration view** in this section displays the Level 2 results next to the HEA actuals so you can tune calibration inputs (such as hot-water use per person or system efficiencies) until the model matches the bills.

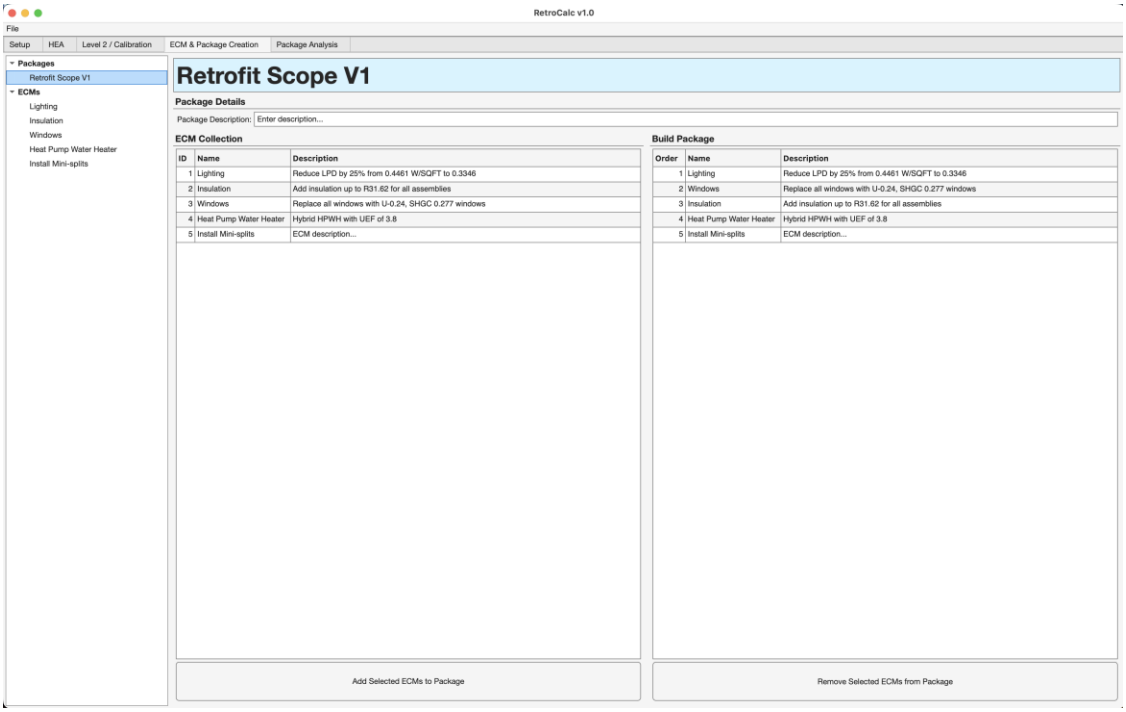


Alongside the built-in appliance categories (stoves, dryers, dishwashers, clothes washers, refrigerators), RetroCalc also supports a Custom Appliance type for anything not covered by the defaults — pool pumps, EV chargers, commercial equipment, dedicated electric heaters. You enter the annual electric and gas use directly, plus the sensible and latent fractions of that energy that become internal gains in the conditioned space. See Section 8.7 for how appliance energy is modeled.

5.4 ECM Package

Here you build your library of measures and assemble packages.

- An **ECM** is a named set of changes to the building model — for example, "raise boiler efficiency to 95%" or "add R-10 wall insulation."
- A **Package** is an ordered list of ECMs that are applied together. Because they apply in order, each ECM's savings reflect the building state left by the ones before it.



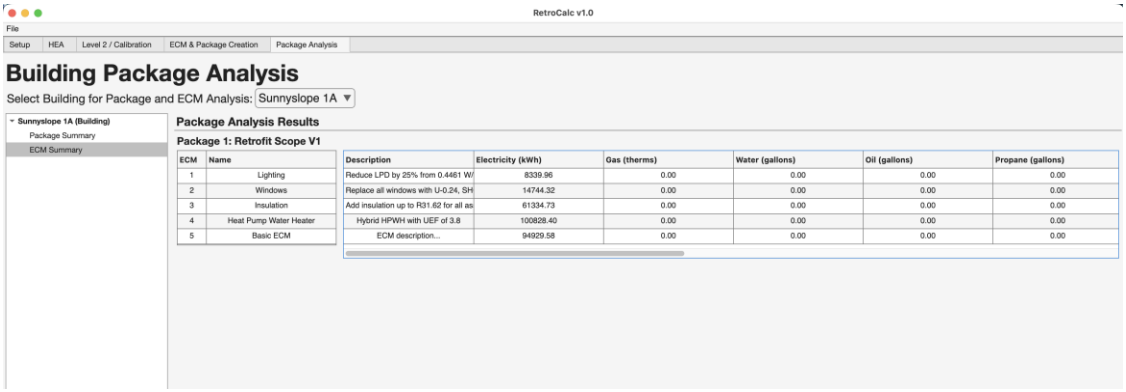
Each measure inside an ECM is one of three operations: Insert (add a new component to the building), Edit (change a property of an existing component), or Delete (remove a component). Edit operations pick the target component by filter criteria — either by matching a property value ("HeatCool where SystemType equals GasBoiler") or by pointing at a specific object index in the component list. Each measure also chooses how its value combines with the existing one: Replace, Add, or Multiply.

For properties that are arrays — like HeatCoolSetpoints, which carries six values (heating and cooling setpoints for day and night, plus humidity limits) — the property picker expands the array into one entry per element. You can target a single element, for example HeatCoolSetpoints(1) — HtgTP1 (day), with Add 2 to raise the daytime heating setpoint by 2 °F without disturbing the other five values.

ECMs you build can be saved to disk with Save ECM and reused on other projects with Load ECM. Each ECM goes to its own file of your choice; loading imports the ECM into the current session's library, skipping any with a name that is already present.

5.5 Package Analysis

Finally, you assign one or more packages to a building and run the analysis. RetroCalc applies each ECM in turn, re-runs the Level 2 model after each one, and reports the cumulative change in energy, cost, and carbon relative to the baseline — both per package and per individual ECM within a package.



Alongside the top-level results, the Export ECM Diagnostics button writes a single Excel workbook containing every intermediate Level 2 table — monthly space-heating and cooling, electric end-use breakdown, heat-loss components, heat-gain components, solar gains by window, DHW energy, appliance results — for the base building and for each applied ECM, one sheet per state. Use this when you need to QC where an ECM's savings come from. The button requires that Apply Packages to Building has been run in the current session; the intermediate per-ECM building states are not preserved across session save/load, so re-apply packages after reloading a saved session before exporting diagnostics.

Part 2 — Reference

This part explains every input in depth and shows how it flows into the calculations.

6. Setup Inputs in Detail

6.1 Site

Field	Description
Project / site name	Free-text label for the project.
Location — city / state	Used to label the project and to look up location-dependent defaults.
Location — latitude / longitude	Used to pull solar and ground-temperature data and historical degree days for the site.
Carbon factors (per fuel)	Kilograms of CO_2-equivalent per unit of each fuel, used to compute emissions.
Shoulder / non-heating months	Months treated as outside the heating season when separating bills into end uses.
Past years to average	Number of recent weather years averaged when normalizing heating and cooling energy.

> **Note.** RetroCalc locates a building by **city/state and latitude/longitude**, not by an ASHRAE climate zone. There is no separate street-address field.

6.2 Building

Field group	Fields
General	Name, year built, number of stories, thermal-mass class (Light / Medium / Heavy)
Units	Total units and the count of 0-, 1-, 2-, and 3-bedroom units
Areas	Gross conditioned area (ft^2) and common/non-residential area (ft^2)
Seasons	Heating (and cooling) season start and end dates
Comfort & schedules	Heating/cooling setpoints (e.g., 72 °F heat / 74 °F cool, 40–60% RH) and HVAC operating schedule (e.g., an occupied period of 6am–6pm with different setpoints for the rest of the day)

The **heating-season dates** matter for the bill analysis: degree days outside the heating season are masked so that summer gas use isn't misread as space heating. **Thermal-mass class** and **setpoints** feed the Level 2 model (see Section 8).

> *Confirm the exact on-screen field labels against the live app when capturing screenshots; the groupings above reflect the model inputs.*

6.3 Utility Meters and End-Use Flags

For every meter you import monthly billing data (period start/end, usage, cost) and set flags that tell RetroCalc **which end uses that meter serves**. The flags drive how the bill analysis splits the meter's consumption.

Electricity meter — 4 end-use flags:

Flag	Meaning
Base load	Meter carries general base/plug load
Space heating	Meter serves electric space heating
Cooling	Meter serves space cooling
DHW	Meter serves electric water heating

Plus a **base-load amplitude** and a **seasonal DHW amplitude** — each is the fractional swing of a winter peak above the summer minimum (e.g., 0.2 means the winter peak is about 20% above the summer-minimum load).

Gas / oil / propane meter — end-use flags:

Flag	Meaning
Space heating	Meter serves space heating
DHW	Meter serves domestic hot water

Flag	Meaning
Cooking	Meter serves cooking appliances (gas/propane)
In-unit clothes dryer	Meter serves in-unit dryers (gas/propane)
Common-area clothes dryer	Meter serves common-area dryers (gas/propane)

Plus a **seasonal DHW amplitude** controlling the monthly hot-water shape (same fractional-swing convention as the electric meter's amplitudes above). (Oil is not a cooking fuel, so it uses only the space-heating and DHW flags.)

Water meter: water meters carry pre-split end uses (residential, irrigation, cooling-tower, other) and a building-allocation ratio rather than end-use flags.

> When one meter serves more than one building, a **meter-to-building ratio** assigns each building its share of that meter's usage.

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7. How the Historical Energy Analysis Works

The HEA turns raw bills into a weather-normalized, end-use-resolved picture. It runs independently of the Level 2 model.

7.1 Separating bills into end uses

For each **electricity** meter, RetroCalc fits a statistical relationship between monthly use and the weather:

```
Monthly electricity  =  Base load
                     +  (Heating slope × Heating degree days)
                     +  (Cooling slope × Cooling degree days)
```

- The **base load** is the weather-independent portion (lights, plugs, always-on equipment).
- The **heating slope** and **cooling slope** are the additional kWh per degree day, isolating the weather-driven heating and cooling.
- Hot water and other flagged uses are carved out using the meter flags and a seasonal shape.

For **gas, oil, and propane**, the same idea applies but **without a cooling term** (these fuels don't run cooling):

```
Monthly fuel  =  Base  +  (Heating slope × Heating degree days)
```

Cooking and clothes-drying energy are **not** derived from the regression. They are estimated from built-in reference data based on the building's unit count and bedroom mix, then subtracted out. Hot water for fuel meters is found using a **summer-minimum** method: the

lowest summer months (when there's no heating) reveal the baseline hot-water load, which is then shaped across the year by the seasonal DHW amplitude. Whatever remains after cooking, drying, and hot water is attributed to space heating.

7.2 Weather normalization

Raw heating and cooling energy depend on how harsh the year was. To make years comparable, RetroCalc rescales the seasonal portion to a typical weather year:

```
Normalized heating energy = Average HDD × Heating slope
```

where **Average HDD** is the multi-year average from the historical degree-day file, anchored to the last billing year and using the number of years you set in Setup. Cooling is normalized the same way with CDD.

7.3 Rolling up

Each meter's results are scaled by its building-allocation ratio and summed to the building level, then buildings are combined (area-weighted) to the site level. The results populate the building's and site's **key results** — see Section 10.

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8. How the Level 2 Model Works

The Level 2 model is a steady-state **monthly energy balance**. The sections below describe each piece and the inputs that control it. Equations are shown in engineering form, not code.

8.1 Envelope heat loss

Every enclosure surface has a **heat-loss coefficient** — its area times its U-value (or, for an opaque surface, area ÷ R-value). RetroCalc sums these across walls, roofs, doors, windows/skylights, below-grade walls and floors, and slab edges to get the building's total conductance. Monthly conductive heat loss is:

```
Conductive heat loss = Σ (surface heat-loss coefficient) × Heating degree
 days × 24
```

Component	Key inputs
Opaque surfaces (walls, roofs, doors)	Area, R-value, orientation
Glazed surfaces (windows, skylights)	Area, U-value, solar heat gain coefficient (SHGC), orientation, monthly shading factor (1 = no shade – 0 = full shade)
Below-grade surfaces	Wall/floor area, depth, insulation
Slab on grade	Perimeter, edge insulation

8.2 Infiltration and ventilation

Infiltration (uncontrolled leakage) is estimated from the building's measured air-tightness (leakage at 50 Pa per ft^2 of envelope), an effective-leakage factor (a dimensionless multiplier on the tightness measurement; default 0.055), the number of stories, and a wind-shielding class (1 = unshielded/exposed – 5 = heavily shielded by nearby buildings or trees; default 3, per ASHRAE Fundamentals shielding classes), combined with monthly wind speed and indoor-outdoor temperature difference. The model converts the tightness measurement into a natural air-change rate using built-in stack-effect and wind-shielding tables.

Ventilation (deliberate mechanical airflow) is computed from each air-mover's airflow, run schedule, and heat-recovery effectiveness. Energy-recovery ventilators reduce the effective conditioning load by their recovery effectiveness; balanced and unbalanced airflows are tracked separately.

Both infiltration and ventilation add to the heating and cooling loads.

8.3 Internal and solar gains

- **Internal gains** come from people, lighting, equipment/plug loads, appliances, and hot-water distribution losses. RetroCalc uses built-in **space-type** data (lighting and equipment power densities, occupancy, and heat-gain rates per person) applied to each space.

- **Solar gains** come through glazing. For each window orientation the model pulls monthly solar data and multiplies by glazed area, SHGC, and a monthly shading factor.

In winter these gains **reduce** the heating load; in summer they **add** to the cooling load.

8.4 Utilization factors

Not all internal and solar gains are useful. A south-facing solar gain at noon in a lightly built building may simply cause overheating rather than offsetting the morning's heating need. RetroCalc applies **utilization factors** (values between 0 and 1) to discount gains that can't be effectively used:

- **Heating** uses the **Barakat–Sander** method (ASHRAE), which depends on the ratio of gains to losses and the building's thermal mass.

- **Cooling** uses the **ISO 13790** monthly method, which depends on a time constant derived from thermal mass and the ratio of gains to losses.

Heavier thermal mass generally improves utilization by buffering swings.

8.5 Net loads and equipment energy

The monthly **net heating load** is the conductive + air loss minus the *utilized* internal and solar gains (never less than zero):

```
Net heating load = Heat loss − (utilized internal gains) − (utilized solar gains)
```

The monthly **net cooling load** adds solar and internal gains to the (utilized) conductive gains.

Net loads are converted to **fuel or electricity** using each system's efficiency, which **varies with outdoor temperature** via a built-in efficiency curve, multiplied by a distribution efficiency (a fraction representing duct/pipe losses, e.g. 0.92 = 8% distribution loss; if left blank RetroCalc fills in a default by system type):

```
Heating fuel input = Net heating load ÷ (seasonal system efficiency × distribution efficiency)
Cooling electricity = Net cooling load ÷ (seasonal COP × distribution efficiency)
```

The model converts to the correct unit per fuel (electricity → kWh, gas → therms, oil and propane → gallons).

System types and the efficiency they use:

Heating type	Efficiency basis	Cooling type	Efficiency basis
Gas boiler / furnace	AFUE	Window AC / PTAC	EER
Air-source heat pump (multi-split, VRF)	Heating COP	Air-source heat pump (multi-split, VRF)	Cooling COP
Electric resistance	COP = 1.0	Water- or ground-source heat pump	Cooling COP
Water- / ground-source heat pump	Heating COP	Air- or water-cooled chiller	COP

8.6 Domestic hot water

DHW energy is built up from demand and losses:

```
Usage energy (per fixture, per month) = Gallons heated × 8.34 × (hot temp − cold supply temp) ÷ 1000  [kBtu]
```

Cold supply temperature varies seasonally (coldest in February, warmest in August). To the usage energy the model adds three loss terms — **tank standby losses**, **mechanical-room pipe losses**, and **recirculation (circulation) losses** — then divides by the water heater's **seasonal efficiency** to get input fuel. Recirculation and pipe losses partly become internal heat gains to the building.

Component	Key inputs
Water heater (DHW system)	Type, fuel, steady-state efficiency, output temperature, recirculation settings (recirculation loop type — CompactInsulated / Average / LongRunsPoorlyInsulated — plus the fraction of fixture load lost to circulation at the summer minimum and its seasonal swing)
Storage tanks	Quantity, volume, standby loss rate, hours hot
Mechanical-room piping	Pipe type, diameter, length, hours hot
Plumbing fixtures	Type, flow per use, uses, use temperature

8.7 Appliances

Each appliance row has a Type (gas or electric stove, in-unit or common-area dryer, refrigerator, dishwasher, clothes washer, or Custom Appliance), a Quantity, an Efficiency Level, and a Fraction of Units Served. For the built-in types, RetroCalc multiplies an energy constant from its reference tables — which already account for average bedroom count — by the Quantity you enter. Each appliance also contributes to internal gains: part of its energy becomes sensible heat in the conditioned space, part becomes latent (moisture), and some leaves the building (vented dryers, for example). The built-in reference tables provide those sensible / latent fractions for the standard appliance types.

If an appliance isn't on the standard list — a pool pump, an EV charger, server rack, commercial oven, or a dedicated heater — choose Custom Appliance from the Type dropdown. For Custom, you enter the annual electricity use (kWh) and annual gas use (therms) directly on the appliance row, along with the sensible and latent gain fractions. Either fuel value can be zero, so the same Custom type handles electric-only, gas-only, and mixed-fuel equipment. Typical gain fractions are 0.9 sensible / 0 latent for an electric appliance in the conditioned space, 0.4 / 0.3 for a gas appliance in-space and unvented, and roughly 0.05 / 0 for anything vented outside.

Quantity is the primary multiplier for every appliance type. For a whole-building appliance (one pool pump, one commercial oven) enter 1; for a per-apartment appliance in a 50-unit building, enter 50. The Fraction of Units Served field is kept for backward compatibility with older projects created before Quantity was wired into the energy math — if Quantity is left at zero, the model falls back to the old units × fraction formula so existing results don't change. New projects should set Quantity and leave the fraction field alone.

9. Built-In Default Data

RetroCalc ships with a set of reference tables that supply default coefficients, efficiency curves, and lookup values. You don't enter these; they're applied automatically, though

several can be overridden in the Level 2 calibration view. Knowing what's behind the numbers helps when calibrating.

Default data	What it provides	Where it's used
Air-leakage rates	Leakage by tightness class (tight / average / leaky)	Infiltration
Stack & shielding factors	Convert 50-Pa tightness to natural air changes	Infiltration
Thermal-mass capacity	Effective thermal mass by construction class	Heating & cooling utilization factors
Ground temperature & soil conductivity	Undisturbed soil temperature and conductivity	Below-grade heat loss
Pipe heat-loss table	Heat loss per foot by pipe type and diameter	DHW distribution losses
Summer-peaking water distribution	Monthly shape for irrigation / cooling-tower water	Water analysis
Stove & dryer data	Annual cooking/drying energy by unit count and bedrooms	HEA and Level 2 appliances
Appliance data	Typical energy by appliance type	Level 2 appliances & internal gains
Space-type data	Lighting/equipment power density, occupancy, heat gains per person	Internal gains
Heating & cooling system data	Default efficiencies and temperature-dependent efficiency curves	Space heating & cooling
Oil heating values	kBtu per gallon by oil grade	Oil energy conversion
Weather station list	City weather stations with latitude/longitude	Weather and degree-day lookup

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One exception: the Custom Appliance type (Section 8.7) does not consult the built-in appliance data table. Its annual energy and gain fractions come entirely from the values you enter on the appliance row.

10. Reading the Results

Both the HEA and the Level 2 model produce a **key-results snapshot** for each building (and the HEA also at the site level). Holding them in the same form lets you compare bills vs. model directly.

Headline results

Result	Units	Meaning

Result	Units	Meaning
Energy use intensity (EUI)	kBtu/ft^2/yr	Total annual energy per conditioned area
Electricity / Gas / Water / Oil / Propane	kWh, therms, gal	Annual consumption by fuel
Annual cost (per fuel and total)	$	Annual utility cost
Carbon (CO_2e)	kg CO_2e/ft^2/yr	Emissions intensity

End-use intensities

Result	Units	Meaning
Space heating	kBtu/ft^2	Heating energy intensity (across fuels)
Space cooling	kBtu/ft^2	Cooling energy intensity
Domestic hot water	kBtu/ft^2	Water-heating intensity
Non-HVAC electricity	kBtu/ft^2	Lights, plugs, fans, pumps
Appliance fuel	kBtu/ft^2	Cooking and drying (excludes hot water)
Water intensity	gal/day/bedroom	Residential water use

Detailed tables. For each fuel, RetroCalc also produces an annual usage table and a monthly profile:

- The **annual usage table** has a *weather-normalized average* row, a *fraction-of-total* row (each end use as a share of the whole), an *intensity* row (kBtu/ft^2), and one row per year of actual billed data.
- The **monthly profile** shows a normalized average month for each end use across the year.

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11. Troubleshooting

Symptom	Likely cause	What to do
A meter's data won't import	Sheet name or column layout doesn't match the expected format	Check the meter sheet against the column lists in Section 4; each meter needs its own sheet.
Heating or cooling looks unnormalized / erratic across years	Degree-day history is missing years or doesn't reach the last billing year	Extend the historical degree-day file to cover all billing years plus the averaging window set in Setup.
Summer gas reads as space heating	Heating-season dates are wrong	Set the building's heating-season start/end so summer months are excluded.

Symptom	Likely cause	What to do
HEA end-use split looks wrong for a meter	End-use flags don't match what the meter actually serves	Review the meter's flags in Setup.
Level 2 doesn't match the bills	Calibration inputs need tuning	Adjust calibration values (hot-water use per person, system efficiencies, etc.) in the Level 2 calibration view until the two agree.
Cooling shows up on a gas building	Cooling flag set on a fuel meter	Only electricity meters carry a cooling flag; clear it elsewhere.

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12. Glossary

- **AFUE** — Annual Fuel Utilization Efficiency; seasonal efficiency rating for fuel-fired heating.
- **Baseline** — The existing building modeled with no measures applied; the reference for all savings.
- **CDD (Cooling Degree Days)** — Accumulated degrees by which daily mean temperature exceeds the base (70 °F here). Higher → more cooling.
- **CO_2e** — Carbon-dioxide-equivalent emissions, in kg.
- **COP** — Coefficient of Performance; heat delivered per unit of energy input (used for heat pumps and chillers).
- **DHW** — Domestic Hot Water.
- **ECM** — Energy Conservation Measure; one retrofit.
- **EER** — Energy Efficiency Ratio; cooling efficiency rating.
- **End use** — A consumption category (base load, heating, cooling, DHW, cooking, drying).
- **EUI** — Energy Use Intensity, kBtu/ft^2/yr.
- **HDD (Heating Degree Days)** — Accumulated degrees by which daily mean temperature falls below the base (65 °F here). Higher → more heating.
- **HEA** — Historical Energy Analysis; the bill-based analysis.
- **Infiltration** — Uncontrolled envelope air leakage.
- **Level 2** — The physics-based engineering model.

- **Package** — An ordered group of ECMs applied together.
- **SHGC** — Solar Heat Gain Coefficient; fraction of solar energy admitted through glazing.
- **Utilization factor** — A 0–1 factor discounting internal/solar gains that cannot be usefully applied.
- **Ventilation** — Deliberate mechanical outdoor-air supply.
- **Weather normalization** — Adjusting seasonal energy to a typical weather year for fair comparison.

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13. Appendix: Scripted and Batch Use (Advanced)

> *This appendix is for advanced users who want to run many buildings automatically. It is optional — the graphical app does everything described in Parts 1 and 2 without any scripting.*

In addition to the graphical app, RetroCalc's engine can be driven from the MATLAB command line, which is useful for batch analysis across many buildings. The same workflow applies: build a site, import bills and weather, run the HEA, run Level 2, then apply packages. A minimal run looks like this:

```
site = ece.Site();
site.importSiteInputsData('MyProject_Site.xlsx');
site.Location.importWeatherData('MyProject_DD.xlsx');
site.ElectricMeters(1).importUsageTable('MyProject_Utility.xlsx', 'Elec_Meter
1');
site.computeBuildingUtilityUsages();   % prepare meter data
site.Buildings(1).computeHEA();        % run the bill-based analysis
site.Buildings(1).computeLevel2();     % run the physics model
```

Results live on each building as `HEA` and `Level2` snapshots (the same numbers shown in the app's results screens). For the full programming interface — every class, property, and method — see the developer documentation in the project's `documentation/codebase-reference/` folder.

Appendix A: Excel Input Reference and API

RetroCalc's UI is where you enter and review data, but three kinds of Excel workbook (`.xlsx`) feed it. This appendix lists what each workbook needs to contain; exact sheet names and cell layouts are described in detail alongside the matching UI fields in Section 6.

File	Purpose	Contents

File	Purpose	Contents
Site input workbook	Buildings and site properties	One row per building with its physical characteristics.
Utility usage workbook	Monthly billing records	One sheet per meter, with a row per billing period.
Historical degree-day file	Weather history for normalization	One row per day with heating and cooling degree days.

Building rows

Column	Description
Name	Building display name
Year built	Year of construction
Number of stories	Above-grade stories
Number of units	Total dwelling units
Bedroom mix	Count of 0-, 1-, 2-, and 3-bedroom units
Gross conditioned area	Conditioned floor area (ft^2)
Common area	Common / non-residential area (ft^2)
Heating season start / end	Dates bracketing the heating season

Utility billing rows (one sheet per meter)

Column	Description
Billing start / end	Dates of the billing period
Usage	Quantity billed (kWh, therms, CCF, or gallons depending on fuel)
Cost	Dollar amount billed

Historical degree-day rows (one row per day, no gaps)

Column	Description
Date	Calendar date
HDD65	Heating degree days, base 65 °F
CDD70	Cooling degree days, base 70 °F

Two of RetroCalc's features call third-party web services for reference data: historical degree-days (degreedays.net) and solar radiation + typical-year weather (NREL PVWatts). Both need API keys supplied as environment variables set before you launch MATLAB. MATLAB reads environment variables only at launch, so if you update them while MATLAB is running, close MATLAB and reopen it.

Degree Days (required if you import HDD/CDD via API)

Used by: the historical degree-days API import on the Site setup screen. Create a free account at degreedays.net and note the account key and security key.

Windows — open Command Prompt (Win+R → cmd → Enter) and run:

setx DEGREEDAYS_ACCOUNT_KEY "your-account-key"

setx DEGREEDAYS_SECURITY_KEY "your-security-key"

macOS / Linux — add these to ~/.zshrc or ~/.bashrc, then open a new shell:

export DEGREEDAYS_ACCOUNT_KEY="your-account-key"

export DEGREEDAYS_SECURITY_KEY="your-security-key"

If either variable is unset, the API call fails immediately with a message asking you to set both DEGREEDAYS_ACCOUNT_KEY and DEGREEDAYS_SECURITY_KEY. Set them and relaunch MATLAB.

NREL PVWatts (required for solar gains in Level 2)

Used by: Level 2's solar-gain calculation, which fetches hourly solar radiation and typical-year weather from NREL every time the model runs. Sign up for a free API key at https://developer.nrel.gov/signup/ — the key is issued instantly on registration.

Windows:

setx NREL_API_KEY "your-nrel-key"

macOS / Linux:

export NREL_API_KEY="your-nrel-key"

If the key is unset, the solar fetch returns HTTP 403 and the Level 2 run falls back to zero solar gain per window, logging a warning. The model will still produce results, but they will understate heating savings from solar gain and overstate cooling loads blocked by shading. Set the key for realistic numbers.

Optional: Endpoint URL Overrides

If either provider changes its endpoint URL, you can override the default without editing source. Create a small JSON file in MATLAB's preferences folder — run disp(prefdir) in MATLAB to see where that is — named retrocalc_endpoints.json, with contents like:

```
{
  "pvwatts_url":    "https://new-host/api/pvwatts/v8.json",
  "degreedays_url": "https://new-host/path"
```

}

The file is read once per MATLAB session and cached. Missing file, missing key, and invalid JSON all fall back silently to the built-in URLs. To pick up a change mid-session without restarting MATLAB, run util.getEndpointOverride("__reload__") at the command line.

Appendix B: HEA Data Collection Checklist

Use this checklist when reaching out to the building owner at project kickoff, well before the site visit. The goal is to have everything you need to run the Historical Energy Analysis before you set foot on site.

Utility bills (per fuel serving the building — electricity, gas, water, oil, propane)

- ☐ 12+ months of billing history per meter, ideally 2–3 years for a stable weather-normalized baseline
- ☐ Billing period start/end dates, usage, and cost for each bill (not just annual totals)
- ☐ Meter numbers/labels if the site has more than one meter per fuel, and which building(s) each meter serves

Building basics

- ☐ Building name(s) and address(es)
- ☐ Year built, number of stories, number of units, bedroom mix (0-, 1-, 2-, 3-bedroom counts)
- ☐ Gross conditioned floor area and common/non-residential area
- ☐ Approximate heating season (and cooling season, if applicable) for the site's climate

Project scope

- ☐ Any retrofit plans or specific ECMs the owner is already considering
- ☐ Planned retrofit timeline, budget constraints, or program requirements (e.g., a funding program's savings targets) that should shape which packages you model

Access

- ☐ Site contact name and availability for a site visit
- ☐ Any access constraints (unit access, mechanical room access, building staff escort requirements)

Appendix C: On-Site Data Collection Checklist

Organized to match the Level 2 model components in Section 8, so each group below maps directly to the inputs you'll enter afterward.

Envelope

- ☐ Opaque surfaces (walls, roofs, doors): area, construction/insulation (or R-value if known), orientation
- ☐ Glazing (windows, skylights): area, type/age (for U-value and SHGC), orientation, shading conditions (nearby buildings, trees, overhangs — shading factor is 1 = no shade, 0 = full shade)
- ☐ Below-grade walls/floors: area, depth below grade, insulation present
- ☐ Slab on grade: perimeter length, edge insulation present

HVAC

- ☐ Heating system(s): type (boiler, furnace, heat pump, electric resistance), fuel, age, nameplate efficiency (AFUE/COP as applicable)
- ☐ Cooling system(s): type (window AC, PTAC, heat pump, chiller), age, nameplate efficiency (EER/COP)
- ☐ Distribution type (ducted, hydronic, etc.) and visible condition/insulation of distribution piping or ductwork, especially runs through unconditioned space

Ventilation & infiltration

- ☐ Mechanical ventilation equipment: type, airflow, run schedule, heat-recovery (ERV/HRV) if present
- ☐ Air-tightness test results if available (blower-door number), or qualitative notes on visible air sealing/leakage
- ☐ General exposure of the building to wind (open site vs. shielded by adjacent buildings/trees) — informs the wind-shielding class

Domestic hot water

- ☐ Water heater(s): type, fuel, age, nameplate efficiency, output/storage temperature
- ☐ Storage tank(s): quantity, volume, approximate standby loss condition (insulation, age)
- ☐ Recirculation: is there a recirculation loop, and does the piping look well-insulated and compact, or long and poorly insulated?

- ☐ Plumbing fixtures: fixture types present (standard vs. low-flow) in units and common areas

Appliances

- ☐ In-unit appliances present: range/oven fuel, clothes dryer fuel, refrigerator age
- ☐ Common-area laundry: count and fuel of washers/dryers

Spaces & occupancy

- ☐ Space types present (residential units, corridors, common areas, mechanical rooms) and their approximate areas
- ☐ Typical occupancy patterns if notably different from a standard residential schedule